Section 2

Thursday, January 22, 2026 11:44 AM

- Quick Review of:
  - Expectation/Variance

    $E(X) = \int x f(x) dx$

    $Var(X) = E(X^2) - E(X)^2$
  - Moment Generating Function (MGF)

    $M_X(t) = E(e^{tX})$

    $M_X(0) = E(e^0) = 1$

    $E(X^k) = M_X^{(k)}(0)$
  - Practice Problems:

    1) $X \sim Exp(\theta)$, $f(x) = \theta e^{-x\theta}$, $x \geq 0$

       a) find the MGF $M_X(t)$

       b) use the MGF to find $E(X)$, $Var(X)$

    2) $X \sim Gamma(\alpha, \beta)$, $f(x) = \frac{\beta^{-\alpha}}{\Gamma\alpha} x^{\alpha-1} e^{-x/\beta}$, $x \geq 0$

       a) "

       b) "

    3) $X \sim Poisson(\lambda)$, $p(x) = \frac{\lambda^x e^{-\lambda}}{x!}$, $x = 0, 1, 2, \ldots$

       a) $M_X(t) = E(e^{tX}) = \sum_{k=0}^{+\infty} e^{tk} \frac{\lambda^k e^{-\lambda}}{k!}$

       $= \sum_{k=0}^{+\infty} \frac{(\lambda e^t)^k e^{-\lambda}}{k!}$

       ↳ $Pois(\lambda e^t)$ Kernel

       $= e^{-\lambda} \cdot e^{\lambda e^t} \sum_{k=0}^{+\infty} \frac{(\lambda e^t)^k e^{-\lambda e^t}}{k!}$

       $$\boxed{M_X(t) = e^{\lambda(e^t - 1)}}$$

       b) $E(X) = \lambda e^{\lambda(e^t-1)} \cdot e^t \Big|_{t=0} = \lambda$

       $E(X^2) = \left[\lambda e^t (e^{\lambda(e^t-1)}) + \lambda^2 e^{2t} e^{\lambda(e^t-1)}\right]\Big|_{t=0}$

       $= \lambda + \lambda^2$

       $Var(X) = \lambda + \lambda^2 - \lambda^2 = \lambda$